INFORMATION TO KEEP WITH YOU

My complete treatment handbook

FEC50

Understand your treatment. Know what to expect. Keep your plan to hand.

Displayed calendar
21 days
See the active treatment phase

Treatment stages
1
Explained in the treatment reference

Regimen source
NCCP 00269
Version 8

> Fluorouracil, epirubicin and cyclophosphamide on Day 1 every 21 days while beneficial and tolerated, within the lifetime anthracycline limit. Radiotherapy must not overlap FEC.

Treatment stages

01 **FEC50 · treatment cycles**
21-day interval

Your team confirms the active stage and dates. The full treatment guide explains each step.

Inside your handbook

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My name _____

Hospital / oncology team _____

> Use the urgent instructions and contact numbers in your treatment guide. Your team must replace example numbers before patient use. Your individual treatment plan takes precedence.

SACTCHECK · CHAPTER 01

Understanding your treatment

FEC50 · National Cancer Control Programme (NCCP) 00269

When to get help

Emergency: call 112 / 999

Severe breathing difficulty, collapse, seizure or severe chest pain: call emergency services. Do not wait for an online answer.

Contact your oncology service immediately

Temperature 37.5°C or higher, 36°C or lower, shivering or suddenly feeling unwell — even with a normal temperature. Seek prompt advice for fever, shivering, severe diarrhoea, painful mouth ulcers, inability to drink, new chest symptoms or a painful/leaking line. Do not wait for the next cycle.

Follow the urgent contact instructions from your treating hospital. If you cannot reach the team, seek urgent medical assessment. Outside clinic hours, use your instructed out-of-hours pathway or the emergency department.

This page does not diagnose symptoms or decide whether it is safe to stay at home. You do not need to know which medicine caused a symptom before calling.

How the medicines work

Treatment to control cancer and symptoms, with regular review of benefit and side effects.

ANTIMETABOLITE CHEMOTHERAPY

Fluorouracil (5-FU)

Short intravenous (IV) injection in FEC



Disrupts deoxyribonucleic acid (the cell's genetic material) (DNA) and RNA production

Fluorouracil interferes with the materials cells need to grow and divide. dihydropyrimidine dehydrogenase (a medicine-processing enzyme) (DPD) testing helps identify people at increased risk of severe toxicity. In FEC it is given on treatment day; it is not a take-home pump infusion.

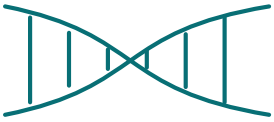
Cell building blocks disrupted → DNA / RNA affected → Growth reduced

What to watch for

Report severe diarrhoea, mouth ulcers or inability to drink promptly. New chest pain needs urgent assessment.

ANTHRACYCLINE CHEMOTHERAPY

Epirubicin



Interferes with DNA handling

Epirubicin interferes with enzymes and DNA processes needed for cell division. It can also affect heart muscle, so the team considers previous exposure and cardiac risk.

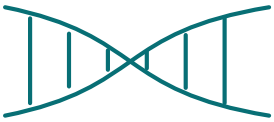
DNA handling disrupted → Cell damage → Division reduced

What to watch for

Report pain or burning at the infusion site immediately. New breathlessness or chest symptoms need prompt review. Red urine for a short time can be a drug-colour effect; pain, clots or persistent bleeding still need advice.

ALKYLATING CHEMOTHERAPY

Cyclophosphamide



Damages DNA needed for growth

Cyclophosphamide is activated in the body and damages DNA. This helps stop dividing cancer cells but can also affect blood-cell production and reproductive tissues.

Drug activated → DNA damage → Cell division affected

What to watch for

Use the prescribed anti-sickness and hydration plan. Report fever, painful urination or blood in the urine.

The drawings are simplified explanations of drug action, not anatomical pictures or a prediction of response.

Why this treatment?

FEC50 may be selected to control metastatic cancer when this chemotherapy combination fits the treatment goal and previous treatment history. It uses a different epirubicin dose from FEC100; the choice depends on benefit, tolerance, heart health and alternatives, not simply a stronger-versus-weaker label.

Your treatment setting

Metastatic breast cancer

The sequence

Fluorouracil, epirubicin and cyclophosphamide on Day 1 every 21 days while beneficial and tolerated, within the lifetime anthracycline limit. Radiotherapy must not overlap FEC.

Alternatives and your priorities

Ask what alternatives fit your cancer, what the treatment aims to achieve, and how its practical demands fit your life. The team can discuss personal benefits and risks.

Why this treatment may be chosen

Show trial results and what the numbers mean

Metastatic breast cancer

- Pembrolizumab + chemotherapy
- Placebo + chemotherapy

Your team can discuss expected benefit and alternatives in the context of your own cancer and priorities.

SACTCHECK · CHAPTER 02

Your schedule, tests and appointments

FEC50 · NCCP 00269

The phases and how they connect

FEC50 · a reference 21-day cycle

21-day view



This shows treatment timing, not guaranteed symptom-free days. Your team confirms the active phase and dates.

21-DAY INTERVAL

FEC50 · treatment cycles

21-day interval

Fluorouracil, epirubicin and cyclophosphamide on Day 1 every 21 days while beneficial and tolerated, within the lifetime anthracycline limit. Radiotherapy must not overlap FEC.

Review scans, symptoms and cumulative exposure; no universal fixed number of cycles is assigned.

BASELINE → TREATMENT → FOLLOW-UP

When checks happen

Relative milestones, not equal time intervals. The clinical team owns booking, result review and action on overdue tests. This is an educational map, not an automated appointment or result tracker.

The team arranges and reviews tests. This guide shows expected timing; it does not record completed appointments. NCCP 00269 v8 · official regimen · ESMO 2024 early breast cancer guideline · treatment and follow-up · ESC 2022 cardio-oncology guideline · cardiac surveillance and risk · NICE NG101 · bone health and follow-up · ESMO metastatic breast cancer guideline · 2026

THE WHOLE TREATMENT JOURNEY

FEC50: treatment and checks

Metastatic breast cancer

Read from left to right. Columns show stages, not equal lengths of time. This is a general plan: no dates or personal information are needed. Your team confirms which steps apply and arranges appointments.

Treatment, tests and reviews across the course

Care / check	Before starting	Early cycles	Later cycles / transition	Next treatment / completion	Follow-up
Cancer medicines NCCP 00269	Agree aim, indication and medicines	FEC50 starts	Review response, toxicity and exposure	Complete planned course / next decision	Ongoing review / subsequent plan
Surgery METASTATIC	multidisciplinary team (MDT)-selected local treatment	No automatic operation during cycles	Review response / surgical plan	Selected for local need	Pathology and recovery guide next steps
Radiotherapy NCCP 00269	Assess indication	Do not give concurrently with FEC	Arrange sequential timing	Radiation team confirms start	Review effects / follow-up
Blood tests NCCP 00269	full blood count (FBC), kidney and liver profile	Review before chemotherapy cycles	Check again at any phase change	Follow active medicines and abnormalities	Follow unresolved abnormalities
Echo / heart NCCP 00269 · ESC	Baseline echo (ESC); assess risk	High cardiac risk: repeat after 2 anthracycline cycles	If continuing: repeat after 4 cycles at high risk	Review cardiac symptoms and total lifetime exposure	Echo within 12 months; high risk within 3 months
electrocardiogram (ECG) NCCP 00269	Baseline ECG in the NCCP anthracycline source; repeat for symptoms or clinical risk. An ECG does not replace an echo.	Symptoms / risk-based repeat	Review if cardiac concern	ECG and echo answer different questions	Clinical indication
Cancer imaging NCCP 00269	Record baseline extent as appropriate	Clinical response / symptoms review	Metastatic: review about2–4 months	Arrange selected surgery / next treatment	Metastatic response plan continues

Care / check	Before starting	Early cycles	Later cycles / transition	Next treatment / completion	Follow-up
DPD / markers / bone NCCP 00269	DPD before 5-FU	No routine tumour-marker clearance	Markers only as selected adjunct in metastatic care	dual-energy X-ray absorptiometry (bone density) (DEXA) if endocrine/ bone-risk indication	Follow the endocrine and bone-health plan
Endocrine / bone BONE · EARLY	Confirm hormone receptors	Plan endocrine therapy when appropriate	Separate from this chemotherapy prescription	Start/continue per MDT; DEXA when indicated	Risk-based bone review

What each check is for

1 Before first dose

Baseline

Confirm DPD testing before first 5-FU; interpret uracil results cautiously in renal impairment. Keep the result available when treatment changes. Baseline FBC, kidney and liver tests; repeat before each chemotherapy cycle.

Ask / act: Confirm tests and consent before prescribing.

2 Heart-monitoring plan

Baseline → cycles → after treatment

ESC supplementary guidance: baseline echo before anthracycline treatment. High/very-high cardiac risk: echo every two anthracycline cycles (for example before cycles 3 and 5 when continuing), and within 3 months after completion. All adults: echo within 12 months after completion. Other repeat tests depend on risk, cumulative exposure and symptoms.

Ask / act: Name the team responsible for booking and reviewing results.

3

Treatment-phase review

Before each transition

Fluorouracil, epirubicin and cyclophosphamide on Day 1 every 21 days while beneficial and tolerated, within the lifetime anthracycline limit. Radiotherapy must not overlap FEC.

Ask / act: Confirm active medicines, support script and the next test checkpoint.

4

Response and follow-up

Planned review points

Record disease sites before treatment. Response imaging is generally considered every 2–4 months under ESMO guidance, adjusted to symptoms and disease pace. For a 21-day course this is approximately after 3–6 cycles; the team chooses the scan and exact checkpoint. New symptoms may need earlier assessment.

Ask / **act:** Review symptoms and imaging together; do not change treatment on tumour markers alone.

At the next review, ask

- Which treatment phase am I in, and what happens next?
- Which test is expected next, who arranges it, and who explains the result?
- Who should I contact if an expected appointment has not arrived?

This map does not show whether a test has been ordered, booked or completed. Do not wait for a scheduled test if you develop new symptoms.

Your practical treatment plan

STEP 1

Before the first visit

Blood tests, relevant cardiac assessment, medicines review, fertility discussion and the consent conversation come first.

STEP 2

Plan venous access

A cannula, peripherally inserted central catheter (PICC) or implanted port may be used. If a port is chosen, a trained team inserts it in a separate procedure with local anaesthetic and sometimes sedation; the team explains fasting, blood-thinner and wound instructions.

STEP 3

Infusion day

Staff check the line, give prescribed premedicines and treatment, and observe for reactions. Tell them immediately about burning, pain, rash or breathlessness.

STEP 4

Home and next review

This regimen does not routinely need a take-home infusion pump. Staff disconnect the infusion; a port is de-accessed when appropriate. Keep line-care and next-visit instructions.

Port insertion and removal

The line team explains placement, dressing care and when normal activities can resume. A port is accessed with a special needle during treatment. Removal of the implanted device is a separate planned procedure when no longer needed; do not remove or adjust it yourself.

PICC care and urgent line symptoms

A PICC usually needs regular dressing and flushing appointments even between treatments. Fever, redness, swelling, pain, leakage or damage requires prompt advice. Your hospital supplies the exact care instructions.

Visit time and transport

Allow time for blood review, preparation, treatment and observation. Your unit confirms the full visit duration. Arrange transport or an escort where advised.

Phase-specific visit pattern

These medicines are given on Day 1 in the unit. No take-home infusion pump is part of this regimen; port needle removal after a visit is different from removal of the implanted port.

Checks across treatment

Your team arranges, reviews and acts on tests; an overdue result needs follow-up.

Blood tests

Baseline FBC, kidney and liver tests; repeat before each chemotherapy cycle.

Heart monitoring

ESC supplementary guidance: baseline echo before anthracycline treatment. High/very-high cardiac risk: echo every two anthracycline cycles (for example before cycles 3 and 5 when continuing), and within 3 months after completion. All adults: echo within 12 months after completion. Other repeat tests depend on risk, cumulative exposure and symptoms.

ECG

Baseline ECG in the NCCP anthracycline source; repeat for symptoms or clinical risk. An ECG does not replace an echo.

Cancer scans

Record disease sites before treatment. Response imaging is generally considered every 2–4 months under ESMO guidance, adjusted to symptoms and disease pace. For a 21-day course this is approximately after 3–6 cycles; the team chooses the scan and exact checkpoint. New symptoms may need earlier assessment.

DPD before 5-FU

Confirm DPD testing before first 5-FU; interpret uracil results cautiously in renal impairment. Keep the result available when treatment changes.

Surgery, radiotherapy and later treatment

In metastatic disease, surgery and radiotherapy are individual decisions for symptoms or selected local-control needs; they are not automatic steps after every course. Concurrent radiotherapy during FEC is contraindicated in the NCCP source; agree a sequential plan.

Hormone-receptor-positive disease

Endocrine treatment is considered when appropriate for the cancer biology and setting. It has its own schedule and monitoring; it is not automatically part of this chemotherapy prescription.

Bone health

DEXA is linked to an aromatase inhibitor, ovarian suppression, treatment-induced menopause or other bone risk—not automatically to every chemotherapy course. The endocrine team confirms baseline testing and risk-based repeat scans.

Markers and follow-up

Tumour markers can support selected metastatic assessments but must not alone determine progression or treatment changes.

SACTCHECK · CHAPTER 03

Symptoms and supportive care

FEC50 · NCCP 00269

FEC50 · a reference 21-day cycle

Day 1 → Day 21: a general pattern

This is a visual orientation, not a prediction of your personal cycle. It shows when some effects often emerge and which effects are **not** tied to a particular day.

Treatment or check	Timing in this view
Treatment · Day 1	 **Day 1** Medicines given in the unit
Nausea / appetite	 **Days 1–5** Often early; can last longer
Blood counts / infection	 **Days 1–21** Timing varies; fever needs immediate advice
Mouth / bowel	 **Days 1–21** Report ulcers, diarrhoea or poor intake
Fatigue	 **Days 1–21** May build over repeated cycles

General orientation: Effects can occur at any time or continue beyond the chart. There are no guaranteed symptom-free days.

No dates to enter: Your team books appointments and reviews results. The graph requires no personal information.

> **Why these bands stay broad:** These broad symptom bands are educational possibilities, not measured risk curves. Infusion days are the planned schedule; delays require a new clinical plan. Band length does not show likelihood or severity, and the chart cannot tell you whether it is safe to receive treatment. NCCP 00269 v8 · official regimen ↗

There are no guaranteed “safe days”. Fever, shivers, sudden illness, severe breathing changes or chest pain need prompt assessment on any day.

Between-treatment notes

Day 1–21: A reference treatment cycle

Fluorouracil, epirubicin and cyclophosphamide on Day 1 every 21 days while beneficial and tolerated, within the lifetime anthracycline limit. Radiotherapy must not overlap FEC. Follow the written appointment and medicine plan; no date entry is needed to use this chart.

Support around the medicines

1 A clear anti-sickness plan

Take the prescribed prevention medicines and keep the rescue instructions. Contact the team if nausea or vomiting prevents drinking.

2 Infection and growth-factor plan

The team decides whether growth-factor injections are needed and gives the exact days. Fever or suddenly feeling unwell needs immediate advice.

3 Hydration and urinary symptoms

Agree fluid intake with your team, especially if you have a heart or kidney fluid restriction. Report painful urination or blood in the urine.

4 Before treatment

Discuss fertility, contraception, other medicines and line access. Tell staff immediately about burning or pain at the infusion site.

5 DPD testing before fluorouracil

Confirm the DPD result before the first treatment. Testing reduces uncertainty but does not rule out all serious side effects.

Questions worth asking

Who arranges the scans and tests?

Your treating service should book the planned checks, review the results and explain any change. If an expected appointment is missing, contact the team; this map does not book tests or confirm that they have happened.

Try explaining the plan back: Can you explain which medicines you are receiving now? What symptoms mean you should call urgently? Which blood tests, scans or heart checks come next, and who arranges them?

Fluorouracil (5-FU)

- Low blood counts and infection
- Diarrhoea and mouth ulcers
- Rare serious heart symptoms
- Severe toxicity with dihydropyrimidine dehydrogenase (a medicine-processing enzyme) (DPD) deficiency

Epirubicin

- Low counts, infection and mouth ulcers
- Nausea, hair loss and fatigue
- Heart damage, sometimes delayed
- Severe tissue injury if the infusion leaks
- Rare later blood cancers

Cyclophosphamide

- Low counts and infection
- Nausea, vomiting and hair loss
- Bladder irritation / bleeding
- Fertility effects and menopause

Supportive prescription draft for review. This is a regimen template. It is not an issued patient prescription; individual review and completion are required.

FEC50 · treatment-day support

Active medicines: Fluorouracil (5-FU) + Epirubicin + Cyclophosphamide

Emetogenic risk: High · anthracycline / cyclophosphamide

Medicine	Dose / route	When
Aprepitant	125 mg orally	Day 1
Ondansetron	16 mg orally	Day 1 Individual dose / QT / hepatic review
Dexamethasone	12 mg orally	Day 1 NCCP anthracycline–cyclophosphamide row: no routine Days 2–4 dexamethasone; local/prescriber review applies.

Medicine	Dose / route	When
Olanzapine	5 mg orally	Days 1–4 Review sedation, falls, interactions and glucose
Aprepitant	80 mg orally daily	Days 2–3
Metoclopramide	10 mg orally up to three times daily	Subsequent days: duration and use to be confirmed Caution with olanzapine: additive extrapyramidal effects; adjust / avoid for individual contraindications

Additional phase-specific support

- Review granulocyte colony-stimulating factor (white-cell growth support) (G-CSF) need with the consultant; FEC count/FN tables include specific alternatives and daily G-CSF Days 4–11 where selected. Pegylated products need their own timing.
- Confirm an individual hydration plan; the source Day 1 fluid advice must be adapted for cardiac or renal restrictions.
- Fertility, contraception, extravasation and urgent-contact advice are part of the treatment discussion.

Before signing: verify allergies, current medicines, interactions, renal / hepatic function, QT risk, sedation, prior response, total steroid exposure and the rescue-medicine duration. Do not duplicate medicines from another phase.

Individual changes / rescue duration: _____
Prescriber / date: _____ Pharmacy review: _____

NCCP 00269 · VERSION 8

FEC50

Fluorouracil, epirubicin and cyclophosphamide on Day 1 every 21 days while beneficial and tolerated, within the lifetime anthracycline limit. Radiotherapy must not overlap FEC.

Source reconciled; independent clinical, pharmacy and patient/PPI review pending. Selected effects; not a complete medicine leaflet.

A picture of side effects

Numbered symptom boxes connect to the anatomical illustration. The medicine responsible may be uncertain; report the symptom.

1 Heart

Epirubicin / fluorouracil

Baseline assessment; report chest pain, breathlessness or swelling.

2 Blood / infection

Chemotherapy phase

Fever, shivers or suddenly unwell: call immediately.

3 Hair / scalp

Chemotherapy phase

Hair loss is common. Ask about scalp cooling before treatment where suitable.

4 Mouth

Chemotherapy phase

Painful ulcers or difficulty drinking need advice.

5 Bladder

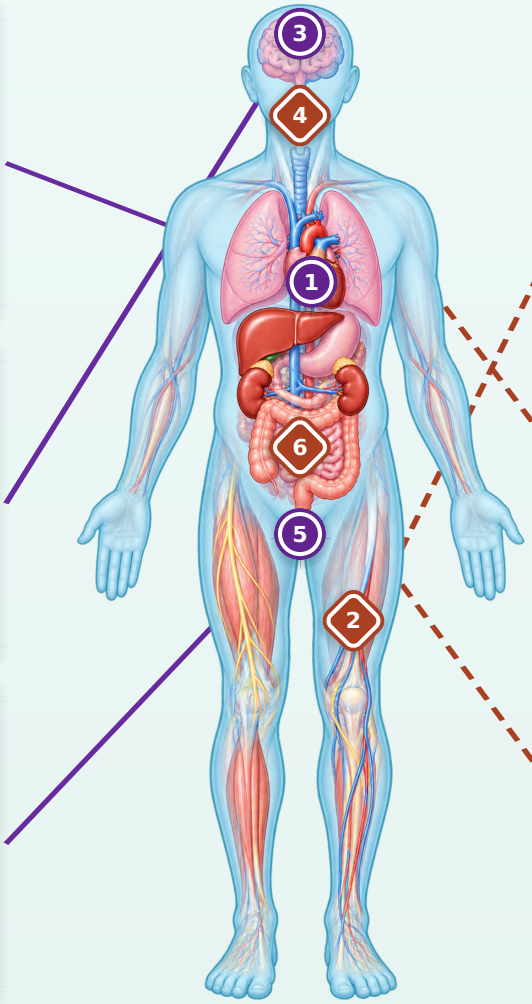
Cyclophosphamide

Report blood in urine or painful urination.

6 Bowel / hydration

Fluorouracil

Diarrhoea, vomiting or inability to drink needs prompt advice.



Selected effects, not every possible effect or a prediction. Blood-count, skin and line markers are schematic. Number and text carry meaning; colour is decorative.

Selected medicine-specific effects are highlighted. The map is not a complete list; persistent or new symptoms need review even after a medicine stops.

> Fever, shivers or suddenly unwell: contact oncology immediately. Severe chest pain, breathing difficulty, collapse or seizure: 112 / 999.

Your record and contact card

FEC50 · National Cancer Control Programme (NCCP) 00269

Record each cycle separately. Each medicine has its own symptom table. Mark the worst symptoms since your last review, when they occurred and what they stopped you doing. Overlapping symptoms can be recorded under more than one medicine; you do not have to identify the cause.

These are patient-language severity prompts, not a validated questionnaire or a formal Common Terminology Criteria for Adverse Events (CTCAE) grading tool. Your team assigns the clinical grade. Serious symptoms override the columns: call without waiting to decide a severity level.

Use a new record sheet for each cycle or review. Copy or print another sheet when needed; the number of sheets does not prescribe the length of treatment.

Cycle number: _____ Start date: _____ Next review: _____

FEC50 · reusable cycle record · copy this sheet for another cycle of the same phase.

Fluorouracil (5-FU) · Antimetabolite chemotherapy

Symptom	None	Mild	Moderate / call	Severe / urgent	Emergency signs
Mouth soreness / ulcers	□ None	□ Mild soreness; eating normally	□ Painful; soft food needed: call	□ Cannot eat / drink enough: urgent review	□ Major breathing difficulty: 112/999
Diarrhoea	□ Usual pattern	□ Up to 3 extra stools above normal	□ 4–6 extra stools/ day or at night: call	□ 7+ extra stools/ day or dehydration: urgent review	□ Collapse / severe dehydration: 112/999
Fatigue	□ Usual energy	□ Tiredness improves with rest	□ Rest does not help; usual tasks affected: call	□ Self-care difficult: urgent review	□ Collapse: 112/999

Epirubicin · Anthracycline chemotherapy

Symptom	None	Mild	Moderate / call	Severe / urgent	Emergency signs
Mouth soreness / ulcers	□ None	□ Mild soreness; eating normally	□ Painful; soft food needed: call	□ Cannot eat / drink enough: urgent review	□ Major breathing difficulty: 112/999
Nausea	□ None	□ Queasy but eating and drinking	□ Intake reduced: contact the team	□ Unable to maintain intake: urgent review	□ Collapse / severe dehydration: 112/999
Fatigue	□ Usual energy	□ Tiredness improves with rest	□ Rest does not help; usual tasks affected: call	□ Self-care difficult: urgent review	□ Collapse: 112/999

Cyclophosphamide · Alkylating chemotherapy

Symptom	None	Mild	Moderate / call	Severe / urgent	Emergency signs
Nausea	□ None	□ Queasy but eating and drinking	□ Intake reduced: contact the team	□ Unable to maintain intake: urgent review	□ Collapse / severe dehydration: 112/999
Fatigue	□ Usual energy	□ Tiredness improves with rest	□ Rest does not help; usual tasks affected: call	□ Self-care difficult: urgent review	□ Collapse: 112/999
Urinary symptoms	□ Usual pattern	□ Mild change: report	□ Pain or frequent urination: call	□ Cannot pass urine / marked bleeding: urgent review	□ Collapse or severe illness: 112/999

Dates / doses missed or changed / symptoms affecting daily life / questions:

Red flags override the table. Fever, shivers, suddenly unwell, chest symptoms or inability to drink: seek prompt advice. Severe breathing difficulty, collapse or severe chest pain: 112 / 999.

Carry your treatment details

Print, cut out and fold. Follow the contact plan from your own hospital.

SACTCHECK · SACT TREATMENT

FEC50

NCCP 00269 v8 · 21-day cycle

Fluorouracil, epirubicin and cyclophosphamide on Day 1 every 21 days while beneficial and tolerated, within the lifetime anthracycline limit. Radiotherapy must not overlap FEC.

1	2	3	4	5	6

Your contact plan

Oncology team: Insert numbers here
Out of hours: Insert numbers here.

Evenings / overnight: instructed out-of-hours pathway or nearest ED.
Emergency: 112 / 999.

Other hospital / own contact: _____

QR: https://sactcheck.com/patient/00269/ · no patient information.

Draft for review · SACTCheck v0.84.17

Sources and further information

FEC50 · National Cancer Control Programme (NCCP) 00269

Use this information alongside the plan agreed with your treatment team. Your team confirms your medicines, doses and appointments.

1. **Regimen selection and source references**
   https://healthservice.hse.ie/documents/6693/269_FEC50.pdf
2. **ESMO 2024 early breast cancer guideline · treatment and follow-up**
   https://www.annalsofoncology.org/article/S0923-7534(23)05104-9/fulltext
3. **ESC 2022 cardio-oncology guideline · cardiac surveillance and risk**
   https://academic.oup.com/eurheartj/article/43/41/4229/6673995
4. **NICE NG101 · bone health and follow-up**
   https://www.nice.org.uk/guidance/ng101/chapter/Recommendations
5. **ESMO metastatic breast cancer guideline · 2026**
   https://www.annalsofoncology.org/article/S0923-7534(26)00902-6/fulltext
6. **eviQ · implanted port patient information**
   https://www.eviq.org.au/p/507
7. **eviQ · peripherally inserted central catheter (PICC) patient information**
   https://www.eviq.org.au/p/543
8. **HSE · chemotherapy side effects and urgent advice**
   https://www2.hse.ie/conditions/chemotherapy/side-effects/
9. **NOGG · section 7, aromatase inhibitors and fracture risk**
   https://www.nogg.org.uk/full-guideline/section-7-strategies-management-osteoporosis-and-fracture-risk
10. **NCCP NCIS antiemetics v8 (2025), Table 1**
    https://healthservice.hse.ie/documents/7180/Supportive_Care_Antiemetics_for_inclusion_NCIS_Medical_Oncology.pdf
11. **ASCO · biomarkers in metastatic breast cancer**
    https://ascopubs.org/doi/10.1200/JCO.2015.61.1459